Automated Wheatgrass Production System – Phase II (Digital Twin) Hardware User Manual

Client Organization: Tardigrade Ltd. and the Foundation of the Energy Collective

Client Name: Sebastien Chabot

Team PB-35

April 2, 2026

Authors: Tahsin Hasan, Hamza Islam, Sam Pan, Ao Sun

ECE Capstone - ELEC/CPEN 491

Change history:

Version	Author	Change(s)	Date
V1.0	Sam Pan	Document created	April 2, 2025

Table of contents

Table of contents **2**
1. Document Purpose and Scope **3**
2. System Overview **4**
3. Intended Users and Hardware Responsibilities **5**
4. Hardware Components **6**
5. Pre-Operation Checks **8**
6. Detailed Maintenance Procedures **9**
7. Normal Hardware Operation **10**
8. User-Level Troubleshooting **11**
8.1 Device Offline Status 11
8.2 No Fresh Sensor Updates 11
8.3 Implausible Temperature or Humidity Readings 11
8.4 Light Reading Does Not Change 11
8.6 Slow Recovery After Restart 12
9. Reference Documents **13**

1. Document Purpose and Scope

This document provides the user-facing operating guidance for the hardware portion of the Automated Wheatgrass Production System - Phase II. Its purpose is to explain, in clear and professional terms, what hardware is present in the deployed monitoring subsystem, how users should verify that it is functioning properly, and what basic actions may be taken when a hardware-related issue is observed. The manual is intentionally narrower in scope than the full system documentation. It focuses on the physical monitoring hardware that supports the digital twin rather than the broader software platform, backend configuration, or design rationale.

The manual applies to the deployed monitoring stack consisting of the ESP32 sensing node, the environmental and light sensors attached to it, the water-flow sensor where installed, the Raspberry Pi backend, and the local network path used to transmit readings to the dashboard. It does not cover mechanical modifications to the conveyor system, advanced maintenance procedures, firmware redevelopment, or stretch-goal features that were outside the core Phase II monitoring scope. Those topics remain covered by the project design, test, and maintenance documents.

2. System Overview

The hardware subsystem extends the existing wheatgrass machine with sensing, communication, and local backend support. In normal operation, the ESP32 periodically reads the connected sensors, packages the readings, and forwards them over the local network to the Raspberry Pi backend. The Raspberry Pi hosts the services that receive, store, and present the data to users through the dashboard. This arrangement allows the physical growing environment to be observed remotely while preserving a simple on-site hardware footprint.

The main hardware role from a user perspective is therefore supportive rather than interactive. Users do not normally operate the sensors directly. Instead, the hardware works in the background so that temperature, humidity, light, and in some builds water-flow activity can be viewed on the dashboard. When the dashboard displays fresh values and the device status is online, this indicates that the key parts of the hardware data path are functioning together as intended.

3. Intended Users and Hardware Responsibilities

This manual is written primarily for operators, capstone evaluators, and other basic users who access the system through the web interface but may occasionally need to confirm that the hardware is powered, connected, and behaving normally. In everyday use, these users are not expected to rewire sensors, modify firmware, or alter backend services. Their role is to observe hardware-related status through the dashboard and perform simple checks such as confirming power, checking visible wiring condition, and restarting the sensing node when appropriate.

More advanced tasks, such as replacing a failed sensor, restoring the Raspberry Pi from backup, or editing the ESP32 configuration, should be handled by a maintainer with access to the corresponding technical documents. This separation is important because the system was designed so that most routine interaction occurs through the GUI website, while direct contact with the hardware is limited to verification, safe handling, and first-response troubleshooting.

4. Hardware Components

The deployed hardware stack is intentionally compact. Each component serves a specific role in sensing, communication, or backend support. Table 1 summarizes the components that users are most likely to see referenced in the dashboard or during basic inspection.

Component	**Primary Function**	**User Relevance**
ESP32 microcontroller node	Reads sensor data and transmits readings over Wi-Fi to the backend.	Must remain powered and connected. If it goes offline, fresh data will stop appearing on the dashboard.
BME280 sensor	Measures temperature and humidity near the growing area.	Supports environmental monitoring and helps users detect abnormal growing conditions.
TSL2591 light sensor	Measures ambient or grow-light intensity.	Used to confirm that lighting conditions are present and changing as expected.
YF-S201 water-flow sensor (if installed)	Detects flow activity during irrigation or misting.	Useful for confirming that watering events are occurring when expected.
Raspberry Pi backend	Runs Home Assistant and related services that receive, store, and display hardware data.	Should remain powered on whenever monitoring is needed.
Local Wi-Fi / hotspot / LAN path	Provides the communication path between the ESP32 and Raspberry Pi.	A network interruption can prevent new readings from reaching the dashboard even when sensors still have power.

Table 1. Hardware components

The hardware design documents describe the same architecture in more technical terms, including the ESP32 sensing node, the environmental and light sensors, the Raspberry Pi backend, and the network link between them. The integration diagram also places these components relative to the growing room, tray area, and sprinkler path, which is useful when physically locating equipment during inspection.

5. Pre-Operation Checks

Before relying on the hardware for monitoring, the user should confirm a small set of pre-operation conditions. First, the system should be used only in the intended indoor environment, with electronics protected from direct spray, standing water, and sustained condensation. Second, the Raspberry Pi backend and the ESP32 node should each have a stable power source. Third, the local network used by the sensing node should be active and reachable. These checks are simple, but they prevent the most common causes of apparent system failure, namely lack of power and loss of connectivity.

Users should also visually confirm that the sensor modules and wiring remain in their expected positions. A shifted light sensor, a disconnected environmental sensor, or a cable that has been strained by movement around the wheatgrass machine may not always produce immediate physical signs, but can lead to missing or implausible readings on the dashboard. A short visual check before operation is therefore appropriate even though the system is designed for automated data collection.

6. Detailed Maintenance Procedures

The recommended startup sequence begins with the Raspberry Pi backend. The Raspberry Pi should be powered first and given enough time to complete its boot process and start the required services. After the backend is ready, the ESP32 sensing node should be powered. Once it boots, it should begin attempting to join the configured network and resume sending data automatically. This sequence helps ensure that the sensing node has an available destination when it starts transmitting.

After startup, the user should open the dashboard and verify three things: the device status appears online, the sensor values are current rather than stale, and the reported readings are plausible for the surrounding environment. In a normal indoor setup, temperature and humidity should remain within a believable room or growing-area range, while the light value should respond to whether the grow light is off, under normal room conditions, or directly illuminated. If a water-flow sensor is installed, flow activity should also appear during watering or misting events. Successful verification at this stage indicates that the sensing, communication, and backend layers are operating together.

7. Normal Hardware Operation

During routine use, the hardware subsystem operates automatically and does not require continuous physical interaction. The ESP32 reads the connected sensors at the configured interval, forwards the data to the backend, and relies on the backend services to store and present the results to the user interface. For this reason, most users will experience the hardware only indirectly through live values, trends, alerts, and device status indicators shown on the dashboard.

Even though direct hardware interaction is minimal, users should understand the normal expectations of the deployed system. The ESP32 and Raspberry Pi should remain powered while monitoring is active. The network path should remain stable enough for readings to reach the backend. Sensor values should update regularly rather than remaining fixed for long periods without explanation. A temporary delay immediately after power-up can be normal, but persistent lack of updates is not. The system was also designed to attempt reconnection when a temporary network interruption occurs, so recovery may happen automatically after short outages.

8. User-Level Troubleshooting

When a hardware-related issue is suspected, the user should begin with safe, visible checks rather than technical reconfiguration. The goal of this section is not to replace the maintenance manual, but to provide a practical first-response procedure that matches the skill level of a typical dashboard user. In most cases, the problem can be narrowed to power, network connectivity, sensor placement, or backend availability.

8.1 Device Offline Status

If the dashboard shows the sensing node as offline, the most likely explanation is loss of power to the ESP32, temporary Wi-Fi interruption, or a failed boot or reconnection attempt. The user should first confirm that the ESP32 remains powered and that the local hotspot or LAN path is active. A brief waiting period may be appropriate after startup or after a network interruption, since the firmware design and test documentation indicate that the node is intended to reconnect automatically when the link returns.

8.2 No Fresh Sensor Updates

If the dashboard remains reachable but the sensor values stop updating, the problem may lie in the communication path or the backend rather than in the sensor itself. The user should confirm that the Raspberry Pi backend is powered and that the dashboard can still be refreshed normally. If the backend appears healthy but data remains stale, the issue should be recorded and escalated for maintainer-level diagnosis.

8.3 Implausible Temperature or Humidity Readings

When temperature or humidity readings appear unreasonable, the user should first consider local physical causes before assuming component failure. The BME280 may have been exposed to moisture, shifted away from its intended position, or affected by a nearby heat source or airflow change. A visual inspection for loose connectors, water exposure, or obvious disturbance is appropriate. If the value remains implausible compared with the room condition, the issue should be referred for maintenance.

8.4 Light Reading Does Not Change

If the light reading remains very low or unchanged over time, the user should check whether the light sensor is obstructed, shadowed, or no longer facing the intended direction. A simple comparison between covered, room-light, and directly illuminated conditions is often enough to

confirm whether the sensor is still responding. If there is no meaningful change despite clear lighting differences, the condition should be documented and escalated.

8.5 No Water-Flow Reading During Watering

In builds that include the optional flow sensor, missing flow data during an expected watering event may indicate either no real water movement or a problem in the sensor path. The user should first verify that watering is actually occurring and that the deployed hardware configuration includes the flow sensor. Because this component may not be present in every build, absence of a flow value is only meaningful when the installation specifically supports it.

8.6 Slow Recovery After Restart

A delayed return of readings after a restart does not always indicate failure. The backend requires time to boot, and the sensing node may require additional time to reconnect and resume transmission. Users should therefore allow a reasonable recovery window after a controlled power cycle before concluding that the system is still offline. If data does not return after that interval, the event should be treated as a persistent fault.

If the issue persists after these checks, the user should stop at the observation stage and refer the problem to the designated maintainer. Tasks such as replacing modules, changing firmware, restoring the backend image, or altering API settings should not be attempted through this manual, because they fall under maintenance and technical configuration rather than routine user operation.

9. Reference Documents

This user manual should be read together with the broader project documentation when additional detail is required. The requirements document defines the monitoring scope and expected system behavior. The hardware design and firmware design documents describe the selected components and communication approach. The sensor and firmware test plans provide the main validation evidence for sensor plausibility, transmission, restart behavior, and reconnection. The hardware maintenance manual should be used when deeper troubleshooting, preventive maintenance, component replacement, or backend recovery is required.

For convenience, the most relevant reference set for hardware users is the hardware design document, firmware design document, sensor test plan, firmware test plan, and hardware maintenance manual. Together, these documents provide the technical depth that was intentionally omitted from this shorter operational manual so that the present document can remain readable for basic users.